Tutorial 0 – Upload MicroPython Firmware

Aim

This tutorial will guide you through the tool used to upload the MicroPython firmware on the Arduino Nano 33 BLE Sense Rev2.

Note

Please ALWAYS use the provided MicroPython installer and firmware (on WebCMS3), as the official installer and firmware do not work as intended.

Do not repeat uploading the firmware once it's been uploaded! This will keep the longevity of the flash memory. Seek help from your tutor if any error happens.

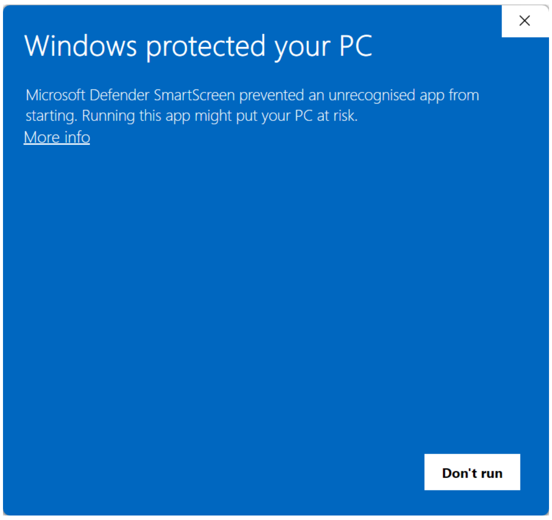
MicroPython Installer

Download the installer from WebCMS3 based on your platform.

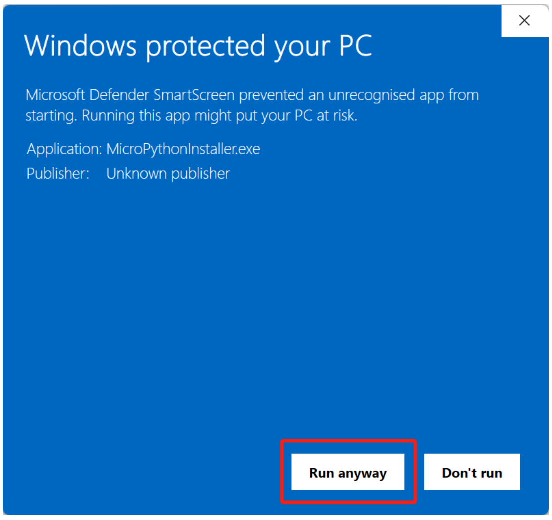
Windows

For Windows users, download "MicroPythonInstaller.exe".

After downloading, double-click the "MicroPythonInstaller.exe". You may get the following warning:



Click "More info" and select "Run anyway".

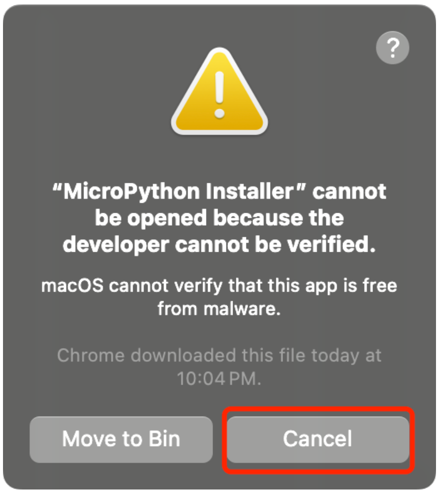


MacOS

For MacOS users, download “MicroPythonInstaller.zip”.

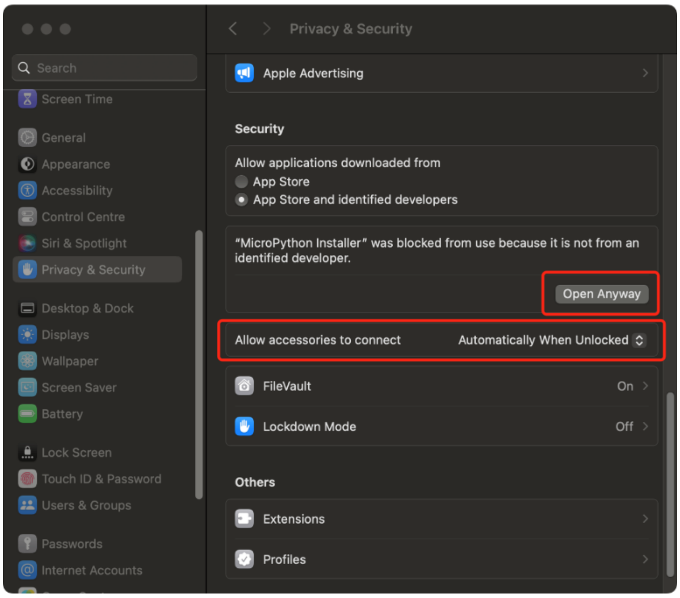
After downloading, extract the MicroPython installer app file depending on your processor (e.g., A/M series should use arm64, and older Mac computers should use x64).

Open the MicroPython Installer app, you may get the following warning:



Click the “Cancel” button and choose Apple menu > System Settings, then click Privacy & Security in the sidebar. (You may need to scroll down.)

First, click the pop-up menu next to “Allow accessories to connect,” then choose “Automatically when unlocked” or "Always". This will allow you to access the Arduino’s serial port.



Then, find the Security section and click “Open Anyway”.

Ubuntu

For Ubuntu users, download “MicroPythonInstaller.deb”.

Open the terminal and change the directory to where the installer is downloaded.

```
sudo dpkg -i MicroPythonInstaller.deb
```

Once the installation is finished, add current user to the dialout group. **Reboot your machine after the command.**

```
sudo usermod -a -G dialout $USER
```

Then you can open the installer by typing:

```
micropython-installer
```

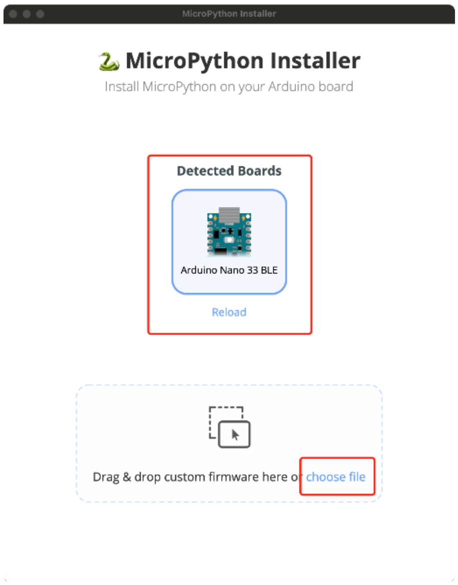
Usage

Download the provided MicroPython firmware from WebCMS3.

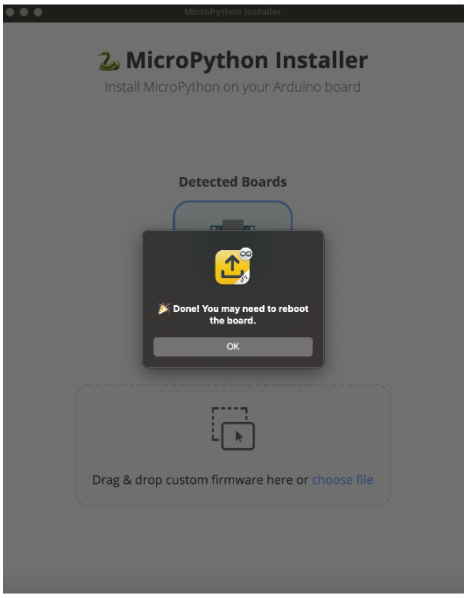
Connect your Arduino with your machine and open the MicroPython Installer.

Double-click Arduino’s reset button to enter the flashing mode. You will see the orange breathing LED.

The installer will detect the board automatically. Click “choose file” and select the downloaded MicroPython firmware.



Once the installer finishes the uploading, you will get the following pop-up window.



Press the reset button on the board and you are good to go!